

1.8 - Impedances

* The main definitions and convention choices of this section were based on [de Sá, PhD Thesis, 2018], while the remarks/properties also took support from the mentioned books of [A.W. Chao], [K.Y. Ng] and [S. Khan].

To further push the power/practicality of the wake formalism - which was up until this point entirely focused on the time/space-domain - we can look at wakes and their effect from a frequency-domain perspective. For this, we introduce the concept of impedances, which are proportional to the Fourier Transforms w.r.t. z of the wake functions:

$$\begin{aligned} Z_{||}(\omega) &= \frac{1}{c} \int_{-\infty}^{\infty} W_{||}(z) e^{i\omega z/c} dz & \text{(I)} \\ \vec{Z}_{\perp}(\omega) &= -\frac{i}{c} \int_{-\infty}^{\infty} \vec{W}_{\perp}(z) e^{i\omega z/c} dz & \text{(II)} \end{aligned}$$

$\hookrightarrow (Z_x, Z_y)$

* In most book ref's you'll find scalar $Z_{\perp}$ and $W_{\perp}$, which is due to considering a cyl. chan
 $\sim \vec{W}_{\perp} = W_{\perp} \cdot \hat{\rho}$

Going back to the more gen. transv. Taylor expansion of $W(\vec{\rho}_s, \vec{\rho}_w, z)$, from which the main wake

terms were $W_0'(z)$ for $W_{||}(z)$, and $W_x^D(z)$, $W_y^D(z)$ and $W^Q(z)$ for $\vec{W}_{\perp}(\vec{\rho}_s, \vec{\rho}_w, z)$, we get

the main impedance terms - $Z_{||}(\omega)$, $Z_x^D(\omega)$, $Z_y^D(\omega)$, $Z^Q(\omega)$ - expressions by making $W_{||}(z) \rightarrow W_0'(z)$ in (I) and $\vec{W}_{\perp}(z) \rightarrow W_x^D(z)$, $W_y^D(z)$ or $W^Q(z)$ in (II), respectively.

Alternatively, we could've defined one generalized impedance:

$$Z(\vec{\rho}_s, \vec{\rho}_w, \omega) = -\frac{i}{c} \int_{-\infty}^{\infty} W(\vec{\rho}_s, \vec{\rho}_w, z) e^{i\omega z/c} dz,$$

and then, expanding $W(\vec{\rho}_s, \vec{\rho}_w, z)$ like before, find all specific impedance terms:

$$Z(\vec{\rho}_s, \vec{\rho}_w, \omega) = \frac{c}{\omega} Z_{||}(\omega) + \overset{\textcircled{\#}}{x_w} \left[\overbrace{Z_x^D(\omega)}^{Z_x} x_s - \frac{1}{2} Z^Q(\omega) x_w \right] + y_w \left[\overbrace{Z_y^D(\omega)}^{Z_y} y_s + \frac{1}{2} Z^Q(\omega) y_w \right]$$

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$$\frac{c}{\omega} Z_{||} = \frac{c}{\omega} \cdot \frac{1}{c} \int_{-\infty}^{\infty} \frac{\partial W_0}{\partial z} e^{i\omega z/c} dz = \frac{1}{\omega} \left[W_0 e^{i\omega z/c} \Big|_{-\infty}^{\infty} - \frac{i\omega}{c} \int_{-\infty}^{\infty} W_0 e^{i\omega z/c} dz \right] = -\frac{i}{c} \int_{-\infty}^{\infty} W_0 e^{i\omega z/c} dz \quad \checkmark$$

Some important remarks:

1. The impedances $Z_{||}$, $\vec{Z}_{\perp}$ have unit of Ohms (Ω) - $\frac{1}{c} \cdot \omega \cdot dz \sim (1/m) \cdot (V/c) \cdot m = \frac{V}{c/s} = \frac{V}{A} = \Omega$ - , analogously to the concept of impedance in electrical engineering. ($Z_{x/y}^D$, Z^Q are in Ω/m)

2. The sign of the exponent in $e^{i\omega z/c}$ is a convention choice, and is equiv. to choosing to write the harmonic component of frequency ω of the beam current (at azimuth s and time t) as

$$I(s, t) = \hat{I} e^{-i\omega(t - s/c)}, \text{ together with } z > 0 \text{ behind the source particle.}$$

To understand this, let's look at how one can naturally get to $Z(\omega)$ when starting from $I(s, t)$:

*Adapted from [K.Y. Ng]

Consider a beam with such current component, $I(s, t) = \hat{I}(\omega) e^{-i\omega(t - s/c)}$. A particle at s will experience

the wake fields from all particles ahead of it, thus with positions $s + z$, and with charges

$$dq(z) = I(s+z, t) \cdot dt = \hat{I} e^{-i\omega(t - s/c - z/c)} \cdot dt = I(s, t) \cdot e^{i\omega z/c} \cdot \frac{dz}{c}.$$

As a consequence, such particle will experience a total voltage - or effective wake potential - given by the sum of all $\rightarrow$ (sometimes normalized by the total charge)

$W_{||}(z) dq(z)$ terms, or better, by the integral of $W_{||}(z)$ times the source charge slices for all

$z > 0$, i.e.:

$$V(s, t) = \int_0^{\infty} W_{||}(z) \cdot \frac{I(s+z, t)}{c} dz = I(s, t) \cdot \frac{1}{c} \int_0^{\infty} W_{||}(z) \cdot e^{i\omega z/c} dz$$

$0 \rightarrow \text{can be set to } -\infty \text{ since } W(z < 0) = 0$

Then, since V is basically a convolution of I and $W_{||}$, it follows that we can also write $V(s, t) = \hat{V}(\omega) e^{-i\omega(t - s/c)}$, and thus:

$$\hat{V}(\omega) e^{-i\omega(t - s/c)} = \hat{I}(\omega) e^{-i\omega(t - s/c)} \cdot \frac{1}{c} \int_{-\infty}^{\infty} W_{||}(z) e^{i\omega z/c} dz,$$

and we get a very natural definition of the long. imp.:

$$Z_{||}(\omega) = \frac{\hat{V}(\omega)}{\hat{I}(\omega)} = \frac{1}{c} \int_{-\infty}^{\infty} W_{||}(z) \cdot e^{+i\omega z/c} dz$$

* Adapted from [S. Khan]

Another - more general - way of arriving at such definition of the impedance is starting with the effective "voltage" felt by a particle due to the wakes of an extended charge distribution, given by the convolution of $w_{||}(z)$ with the charge distribution $\lambda(z)$:

$$V_{||}(z) = \int_{-\infty}^{\infty} w_{||}(z') \cdot \lambda(z-z') dz' = \frac{1}{c} \int_{-\infty}^{\infty} w_{||}(z') \cdot I(z-z') dz',$$

$\lambda \sim C/m = c/v_m \cdot q/m \Rightarrow \lambda = I/c$

~ now adopting the synchronous position as the origin of the long. coord., so everything can be written in terms of z

and then expressing $V_{||}(z)$ and $I(z)$ in terms of their Fourier Transforms:

$$V_{||}(z) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \hat{V}_{||}(\omega) \cdot e^{-i\omega z/c} d\omega; \quad I(z) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \hat{I}(\omega) e^{-i\omega z/c} d\omega$$

$\rightarrow \mathcal{F}^{-1}\{\hat{V}_{||}(\omega)\}$

Convention choice:

- $F(\omega) = \int_{-\infty}^{\infty} f(t) e^{i\omega t} dt$
- $f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega) e^{-i\omega t} d\omega$

implies:

$$\begin{aligned} \frac{1}{2\pi} \int_{-\infty}^{\infty} \hat{V}_{||}(\omega) e^{-i\omega z/c} d\omega &= \frac{1}{2\pi} \cdot \frac{1}{c} \int_{-\infty}^{\infty} w_{||}(z') \cdot \int_{-\infty}^{\infty} \hat{I}(\omega) e^{-i\omega(\frac{z}{c} - \frac{z'}{c})} d\omega dz' = \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} \hat{I}(\omega) e^{-i\omega z/c} \cdot \left(\frac{1}{c} \int_{-\infty}^{\infty} w_{||}(z') e^{+i\omega z'/c} dz' \right) d\omega \end{aligned}$$

Then, defining $Z_{||}(\omega) = \frac{1}{c} \int_{-\infty}^{\infty} w_{||}(z) e^{i\omega z/c} dz$, we get: $\hat{V}_{||}(\omega) = Z_{||}(\omega) \hat{I}(\omega)$, as intended!

* This latter expression is highly important for real-world applications because, in practice, num. simulations enable directly calculating the effective potentials $V(z)$, but not the wake functions $w(z)$ themselves, which are defined for point charges. Then, with the knowledge of $V(z)$ and $I(z)$, one can go to the freq. domain, calculate $Z(\omega) = \frac{\hat{V}(\omega)}{\hat{I}(\omega)}$, and then recover the actual $w(z)$'s through the inverse transform.

3. The additional $-i$ multiplicative factor in the transverse impedance terms is also an usual convention choice, and carries the purpose of making them look more like an ordinary impedance - where the real part is related to the dissipative/growth-damping effects.

↳ Longitudinal wake $\rightarrow$ acts on the dir. of motion $\rightarrow$ naturally performs work

$\rightarrow$ resulting directly in an energy dissipation $\sim \langle P \rangle \propto |\hat{I}|^2 \operatorname{Re}\{Z_{||}(\omega)\}$

↳ Transverse wake $\rightarrow$ acts perp. to the dir. of motion $\rightarrow$ generates a transverse kick $\Delta p \sim \Delta \theta$

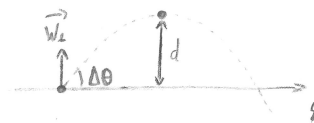
$\rightarrow$ such kick will be translated into transverse displacement only after a $\pi/2$ phase advance (betatron motion) $\rightarrow$ a factor $\pm i$ is needed to put the "dissipative" part of the wake in its real component
 $\hookrightarrow$ which enables work

$\Rightarrow$ With such convention we assure that both $Z_{||}$

and $Z_{\perp}$ align with the standard physical meaning

of impedances, with $\operatorname{Re}\{Z\}$ being the part associated with

irreversible exchanges - net long. energy loss for $Z_{||}$, growth or damping of coherent transverse oscillations for $Z_{\perp}$ - while $\operatorname{Im}\{Z\}$ is the "reactive" part - mainly energy redistribution ($Z_{||}$) or coherent tune shifts ($Z_{\perp}$).



Furthermore, such convention leads to a "cleaner" freq-domain version of the Panofsky - Wenzel

Theorem:

$\nearrow$ considers more terms of $\frac{\partial}{\partial z} W$ (not just W_0')

$$Z_{||}^{\text{full}}(\omega) = \frac{1}{c} \int_{-\infty}^{\infty} w_{||}(z) e^{i\omega z/c} dz \quad \Rightarrow \quad \vec{\nabla}_{\perp} Z_{||}^{\text{full}}(\omega) = \frac{1}{c} \int_{-\infty}^{\infty} \vec{\nabla}_{\perp} w_{||}(z) e^{i\omega z/c} dz \stackrel{\text{P.W.}}{=} \frac{1}{c} \int_{-\infty}^{\infty} \frac{\partial \vec{w}_{\perp}}{\partial z} e^{i\omega z/c} dz$$

$$\Rightarrow \vec{\nabla}_{\perp} Z_{||}^{\text{full}}(\omega) = \frac{1}{c} \left(\left[\vec{w}_{\perp}(z) e^{i\omega z/c} \right]_{z=-\infty}^{z=\infty} - \frac{i\omega}{c} \int_{-\infty}^{\infty} \vec{w}_{\perp}(z) e^{i\omega z/c} dz \right)$$

$$\therefore \boxed{\vec{Z}_{\perp} = \frac{c}{\omega} \vec{\nabla}_{\perp} Z_{||}^{\text{full}}}$$

$$= (W_x^D x_s - W_x^Q x_w, W_y^D y_s + W_y^Q y_w)$$

Resonator:

$$V \uparrow \oplus \approx R \approx L = C$$

RLC circuit in parallel

$$\frac{1}{Z_{||}} = \frac{1}{R_s} + \frac{i}{\omega L} - i\omega C \Rightarrow Z_{||} = \frac{R_s}{1 + iQ\left(\frac{\omega R_s}{\omega} - \frac{\omega}{\omega_R}\right)}$$

where

$$\begin{cases} R_s \sim \text{shunt resistance} \\ Q = R_s \sqrt{L/C} \sim \text{quality factor} \\ \omega_R = 1/\sqrt{LC} \sim \text{res. freq} \end{cases}$$

$$Z_t = \frac{\omega_R}{\omega} \frac{R_s}{1 + iQ\left(\frac{\omega R_s}{\omega} - \frac{\omega}{\omega_R}\right)}$$

Complement on P.W. for impedances

As we've seen, P.W. is a natural "consequence" of writing $w_{||} = \frac{\partial}{\partial z} W$ and $\vec{w}_\perp = \vec{\nabla}_\perp \cdot W$. Thus,

it is also incorporated when using the generalized impedance $Z = -\frac{i}{c} \int_{-\infty}^{\infty} W e^{i\omega z/c} dz$, together with the

definitions $Z_{||}^{\text{full}} = \frac{1}{c} \int_{-\infty}^{\infty} \frac{\partial}{\partial z} W e^{i\omega z/c} dz$ and $Z^{D/Q} = -\frac{i}{c} \int_{-\infty}^{\infty} W^{D/Q} e^{i\omega z/c} dz$, leading to:

$$Z_{||}^{\text{full}} \approx \frac{1}{c} \int_{-\infty}^{\infty} \left[W'_0 + x_w \left(W_x^{D'} x_s - \frac{1}{2} W^{Q'} x_w \right) + y_w(\dots) \right] e^{i\omega z/c} dz = Z_{||} + x_w \left(\frac{1}{c} \int_{-\infty}^{\infty} W_x^{D'} e^{i\omega z/c} dz \right) x_s - \dots = Z_{||} + x_w \left(Z_{x,||}^D x_s - \frac{1}{2} Z_{||}^Q x_w \right) + \dots$$

but also:

$$Z_{||}^{\text{full}} \approx \frac{\omega}{c} Z = Z_{||} + x_w \left(\frac{\omega}{c} Z_x^D x_s - \frac{1}{2} \frac{\omega}{c} Z_{||}^Q x_w \right) + \dots$$

or $\vec{Z}_\perp = \vec{\nabla}_\perp Z = (Z_x^D x_s - Z_{||}^Q x_w, Z_y^D y_s + Z_{||}^Q y_w)$

We could call these: $Z_{x,\perp}^D, Z_\perp^Q$

$$\begin{cases} Z_{x/y,\perp}^D = \frac{c}{\omega} Z_{x/y,||}^D \\ Z_\perp^Q = \frac{c}{\omega} Z_{||}^Q \end{cases}$$